Unraveling the Spectro-polarimetric Difference of Repeating and Non-repeating Fast Radio Bursts

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Abstract

Based on current observational data, Fast Radio Bursts (FRBs) are divided into repeating and apparently non-repeating populations, whose physical relationship remains an open question. Previous studies have mainly compared spectral properties or polarization statistics separately. In this work, we propose a multimodal deep learning framework that jointly learns from dynamic spectra and polarization angle (PA) evolution. Dynamic spectra and PA sequences are encoded by two independent 1D convolutional encoders and fused into a shared latent representation. Unsupervised visualization and supervised classification demonstrate that the learned multimodal features effectively distinguish repeating and non-repeating FRBs. The results suggest that combining spectral morphology with polarization evolution provides complementary information for studying the physical differences between FRB populations.

Introduction

Fast radio bursts (FRBs) are bright millisecond-duration radio bursts of unknown physical origin observed at extragalactic distances (Lorimer et al., 2007; Petroff et al., 2019). Based on current observational data, fast radio bursts (FRBs) are phenomenologically classified into two categories: **repeating and non-repeating**.

Previous studies have compared the two populations using spectral properties (Pleunis et al., 2021), spectral-derived physical parameters (Luo et al., 2023), and more recently deep-learning-based representations extracted from dynamic spectra (Yang et al., 2023). At the same time, polarization observations have revealed valuable information about the emission geometry and magnetized environments of FRBs through quantities such as the polarization fraction, rotation measure (RM), and PA evolution (Pleunis et al., 2021). However, **spectral and polarization information are usually analyzed separately**, and their complementary information has rarely been exploited within a unified learning framework.

In this work, we propose a multimodal deep learning framework that jointly learns from dynamic spectra and PA evolution. By integrating these two observables, we aim to explore whether their complementary information can better characterize the intrinsic differences between repeating and non-repeating FRBs.

Dataset & Data preparation

The dataset consists of 536 bursts from FRB 20201124A, 700 bursts from FRB 20220912A observed by FAST, and 64 apparently non-repeating FRBs from the first CHIME/FRB Baseband Catalog. For each burst, the dynamic spectrum and PA evolution were extracted. Both modalities were resampled to 64 time bins to normalize burst duration and enable the model to focus on spectral morphology and PA evolution rather than absolute temporal scales. To remove the artificial discontinuity caused by the 180° ambiguity of PA, the PA sequence was encoded as $(\cos 2\theta, \sin 2\theta)$ before being fed into the network.

Model and Modelling

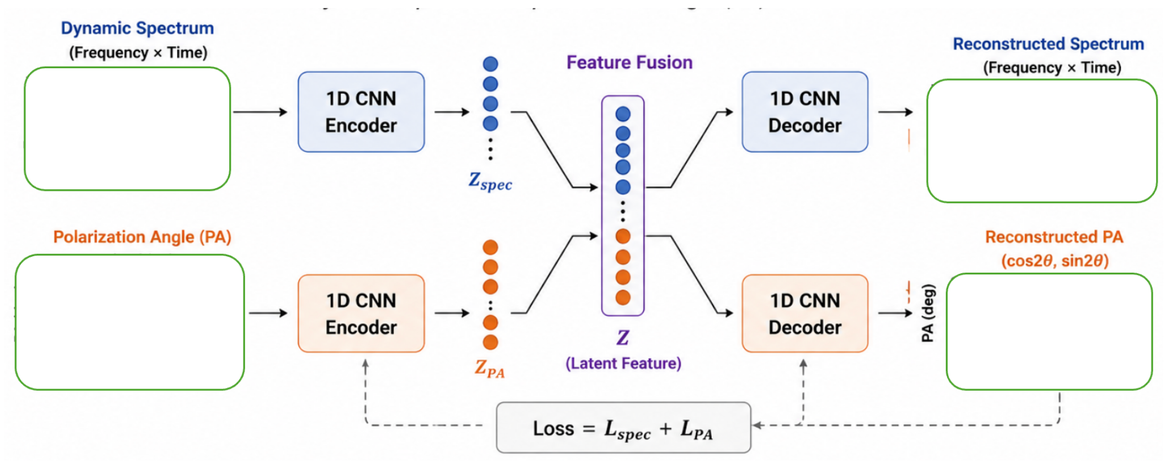


Figure 1. The 256 * 64 dynamic spectra and PA evolution are fed into two independent 1D convolutional encoders to extract modality-specific latent features. The two latent representations are concatenated into a shared feature space and subsequently decoded to reconstruct both modalities. The network is optimized by minimizing the combined reconstruction loss of the dynamic spectrum and PA branches.

The multimodal autoencoder jointly learn the dynamic spectrum and PA evolution of FRBs (Figure: 1). The dynamic spectrum and PA sequence were processed by two independent 1D convolutional encoders, producing modality-specific latent features.

To reduce the influence of different observing frequency bands and RFI environment between FAST and CHIME, the spectral encoder performs convolution only along the temporal dimension, while treating frequency channels as independent input channels. For the PA branch, the PA sequence is encoded as $(\cos 2\theta, \sin 2\theta)$ to eliminate the discontinuity caused by the 180° ambiguity.

The latent features extracted from the two modalities are concatenated into a shared representation, which is then decoded to reconstruct both the dynamic spectrum and the PA sequence. The network is trained by minimizing the sum of the reconstruction losses from the two branches, encouraging the latent representation to preserve complementary information from both modalities.

Results

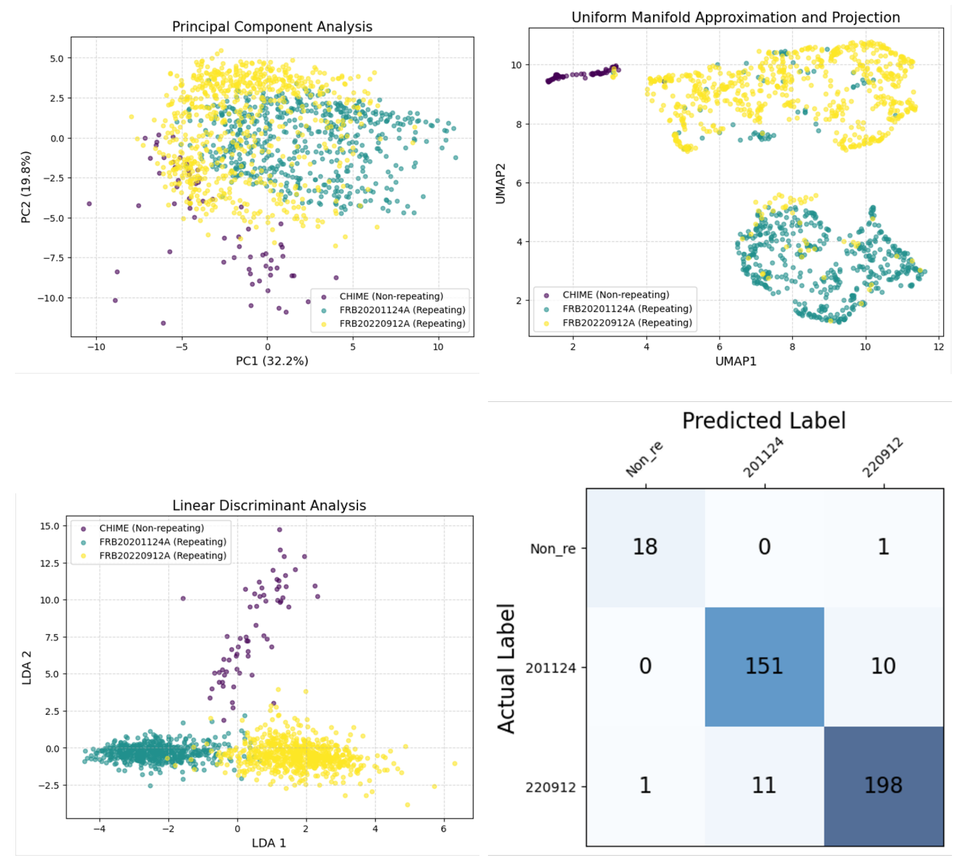


Figure 2. The PCA, UMAP and LDA embeddings of 1300 learned latent representations, and they are respectively located in the upper left, upper right and lower left. The figure in the lower right is the confusion matrix of the XGBoost classification. In the first three pictures, the purple, green and yellow dots respectively represent the non-repeating bursts of CHIME and the repeating bursts of FRB 20201124 and FRB 20220912 observed by fast.

The learned latent representations were evaluated using both unsupervised and supervised methods. Principal Component Analysis (PCA) scikit-learn.org, Uniform Manifold Approximation and Projection (UMAP) umap-learn.readthedocs.io, and Linear Discriminant Analysis (LDA) were applied to visualize the latent feature space, followed by XGBoost classification to quantify the discriminative capability of the learned representations (Figure: 2). PCA reveals limited linear separability, UMAP yields well-defined clusters separating non-repeaters from repeaters, and LDA confirms strong class discrimination. XGBoost achieves 94% accuracy, with some misclassifications between repeaters, indicating that repeaters share common features and non-repeaters are distinct.

Discussion and Summary

Our work combines spectral information with temporal polarization information—specifically, the PA curve—to construct a two-channel one-dimensional convolutional neural network that extracts features from each of these sources separately. We then apply four machine learning methods, including PCA, to analyze the data. The results show that non-repeater bursts exhibit a relatively independent feature space compared to repeater bursts, while different repeater burst sources exhibit significant feature similarity. This indicates that the model has, to some extent, learned the common features of repeating bursts as a class of celestial objects, and also demonstrates that combining polarization and spectral data can enhance the distinguishability between different categories.

Of course, this work still has certain limitations. First, the current sample of repeating bursts consists of only two FRB sources, so **the data set is still relatively limited**. Furthermore, the repeating bursts and non-repeating bursts were observed by different telescopes. Although we minimized the impact of instrumental factors—such as observation frequency and RFI conditions through our network design, **future studies will need to incorporate more samples of repeating and non-repeating bursts from different telescopes** to further verify that the model has learned the intrinsic physical characteristics of FRBs themselves, rather than differences in the observational systems between telescopes. Furthermore, the latent features extracted by the CNN still **lack a clear physical interpretation**; in the future, we hope to conduct further analysis by integrating explainable artificial intelligence.

References

Lorimer, D. R., Bailes, M., McLaughlin, M. A., Narkevic, D. J., & Crawford, F. 2007, Science, 318, 777

Luo, J.-W., Zhu-Ge, J.-M., & Zhang, B. 2023, MNRAS, 518, 1629

Petroff, E., Hessels, J. W. T., & Lorimer, D. R. 2019, A&A Rev., 27, 4

Pleunis, Z., Good, D. C., Kaspi, V. M., et al. 2021, ApJ, 923, 1

Yang, X., Zhang, S.-B., Wang, J.-S., & Wu, X.-F. 2023, MNRAS, 522, 4342

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